

# Jose Manuel Hernandez de Labra

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## EDUCATION

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| <b>B.Sc. in Biomedical Engineering with a Minor in Neuroscience</b><br><i>Tecnologico de Monterrey (ITESM)</i>                                                      | Monterrey, NL, Mexico<br>Aug. 2018 – Dec 2023 |
| <b>M.Sc. in Bioinformatics for Health Sciences</b><br><i>Jointly organized by <u>Universitat Pompeu Fabra (UPF)</u> and the <u>University of Barcelona (UB)</u></i> | Barcelona, Spain<br>Sep. 2025 – Present       |

## EXPERIENCE

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| <b>Neuroscience Department, Zambrano Hospital</b><br><i>Data Scientist Intern</i>                                                                                                                                                                                                                                                                                            | Aug. 2022 – Dec. 2022<br>Monterrey, NL |
| <ul style="list-style-type: none"><li>Cleaned and processed research data collected by psychologists and psychiatrists.</li><li>Developed interactive data visualizations, enabling better comprehension of study findings.</li><li>Applied machine learning models for patient classification and anomaly detection, improving diagnostic precision.</li></ul>              |                                        |
| <b>Core Lab Genomics</b><br><i>Machine Learning Researcher</i>                                                                                                                                                                                                                                                                                                               | Jan. 2024 – May. 2025<br>Monterrey, NL |
| <ul style="list-style-type: none"><li>Designed and optimized machine learning architectures for RNA and biomolecule analysis.</li><li>Predicted 3D RNA structures, contributing to advancements in molecular interaction studies.</li><li>Applied advanced docking algorithms to predict biomolecule-protein interactions, aiding in therapeutic target discovery.</li></ul> |                                        |

## RESEARCH PROJECTS

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| <b>rnAIR</b>   <i>Python, Tensorflow, Molecular Biology, Mathematical Modeling</i><br><i>IGEM competition</i>                                                                                                                                                                                                                                                                                                         | Jan. 2022 – Dec. 2022 |
| <ul style="list-style-type: none"><li>Developed neural network models to predict small RNA binding sites, enabling bacterial gene inhibition.</li><li>Collected and analyzed extensive datasets of small RNAs, addressing antibiotic resistance issues.</li><li>Contributed to mathematical modeling and programming efforts, ensuring model robustness and accuracy.</li></ul>                                       |                       |
| <b>ADA</b>   <i>Python, Docker, Tensorflow, Structural Bioinformatics, Molecular Dynamics</i><br><i>IGEM Design League</i>                                                                                                                                                                                                                                                                                            | Jan. 2023 – Dec. 2023 |
| <ul style="list-style-type: none"><li>Led mathematical modeling and programming efforts to develop biosensors for Alzheimer's detection.</li><li>Designed aptamers using 3D RNA models to bind specific proteins.</li><li>Trained deep learning architectures, including mesh convolutions and GAN's, on Azure Virtual Machines.</li></ul>                                                                            |                       |
| <b>STRAND</b>   <i>Python, Azure, Pytorch, Molecular Dynamics</i><br><i>Tecnologico de Monterrey in collaboration with Microsoft AI For Good Lab</i>                                                                                                                                                                                                                                                                  | Jan. 2024 – Jan. 2025 |
| <ul style="list-style-type: none"><li>Conducted research to improve RNA 3D structure prediction, enhancing insights into RNA folding and interactions.</li><li>Generated labeled datasets with precise intramolecular bindings.</li><li>Developed graph neural networks for improved secondary structure contact maps.</li><li>Utilized contact maps as templates for accurate 3D RNA structure prediction.</li></ul> |                       |

## TECHNICAL SKILLS

**Programming Languages:** Python, Assembler, Bash, SQL, C, C++, LaTeX  
**Developer Tools:** Git, Docker, Azure, VS Code, Visual Studio, Conda, Linux  
**Libraries:** pandas, NumPy, Matplotlib, Tensorflow, Pytorch, SkLearn,  
**Languages:** Spanish - (Native), **English** - (C1), **French** - (A1)

## ADDITIONAL INFORMATION

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| <b>Awards and Achievements</b>                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"><li><b>Two Gold Medals, iGEM Competition (2022, 2023)</b> – Recognized for excellence in synthetic biology research and innovative solutions.</li><li><b>Judge, iGEM Competition (2024, 2025)</b> – Selected to evaluate and mentor competing teams, showcasing leadership and subject matter expertise.</li></ul> |